



KENYA METHODIST UNIVERSITY
DATABASE SYSTEMS CAT BBIT 232/ CISY 221 (40 Marks)

• **ANSWER ALL QUESTIONS**

Question One

a) Discuss the following database related terms: (3 Marks)

- i. Relationships
- ii. Column/ field
- iii. Row/ tuple

b) What are the FOUR areas where databases are found/used? (4 Marks)

c) Write SQL code that generates the table below (5 Marks)

address_id	street	city	state	zip_code
1	807 Grandrose Ave.	Yonkers	NY	10701
2	26 Market Drive	Forest Hills	NY	11375
3	60 Myers Dr.	Amityville	NY	11701
4	9782 Indian Spring Lane	Harlingen	TX	78552

d) What is the use of SELECT statement in DBMS? Give the syntax of SELECT statement and explain the various options (4marks)

e) Discuss FIVE functions of a database administrator. (5 Mark)

f) Write SQL statements to perform the following operations

Customer table

P.D	LAST Name	First Name	Address	City	Salary
1	Hanset	Olap	345	Thika	20,000
2	Standson	Tore	346	Nairobi	30,000

3	Pettarson	Kare	347	Nairobi	40,000
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- i. Displays all customers and all their details from the above table (1 Mark)
- ii. Retrieves last name and first name where city is Nairobi (3 Marks)
- iii. Fetches the different cities available (3 Marks)
- iv. Fetches the count of employees from Nairobi (3 Marks)
- v. Finds the maximum, minimum, total and average salary of the employees. (3 Marks)
- vi. Insert a row with (4, Mike, John, 243, Embu, 50,000). (3 Marks)
- vii. Returns the employee id, last name and full name of employees whose salary lies in the range of 30000 and 60000. (3 Marks)